

Same Signals, Different Actions: Recovering and Validating Investor-Type Preferences from Daily Flow via Inverse Reinforcement Learning

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Abstract

Foreign, institutional, and retail investors trade the same asset for different reasons, yet the regularities that describe them have been established one phenomenon at a time, leaving no common scale on which their preferences can be compared. We build such a scale with inverse reinforcement learning: each type is an agent whose daily net-buy intensity reveals a latent reward over shared market-state features, and all three are estimated under one shared specification, so that differences come only from what is learned. Under a one-step horizon this estimator coincides with a Lasso; our contribution is therefore a validation protocol rather than a new estimator—out-of-sample stability under purged cross-validation and a block bootstrap, construct-validity checks against established behavioral regularities, and explicit disclosure of null results. On 973 trading days of Samsung Electronics (005930) flow, we find that foreign and retail flow load with opposite signs on momentum, each sign-consistent across all 45 out-of-sample splits, while institutional flow reveals no distinct tendency. Applied unchanged to a second stock, Hyundai Motor (005380), the foreign and retail direction partly persists while institutional flow again stays near chance.

CCS Concepts

• Computing methodologies → Inverse reinforcement learning; • General and reference → Validation; • Applied computing → Economics.

Keywords

Inverse reinforcement learning, investor behavior, order flow, behavioral finance, interpretability, model validation

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1 Introduction

On the same day, in the same stock, and facing the same public information, investors act differently. Which conditions tilt each investor type toward buying or toward selling? This is a basic question for price formation, for market microstructure, and for investor-protection policy, since type-level flow is where behavioral tendencies reach the market. The problem we address is how to recover, from observed daily flow alone, a description of what each investor type responds to—and to recover it in a form that can be compared across types.

Existing evidence does not deliver that comparison. The stylized regularities of type-specific behavior—momentum versus contrarian trading [2], the disposition effect [9, 10], institutional herding [7]—have been established one phenomenon at a time, in separate reduced-form regressions on different samples and specifications; of the third we operationalize only the own-following component [14], for the reason given in Section 3.2. Asset demand systems [6] do estimate type-level preferences within a single framework, but from quarterly holdings rather than daily flow, and without behavioral state variables. As a result, even a simple comparison—does foreign flow load more strongly on momentum than retail flow does?—has not been answered under a shared specification.

We approach this problem through inverse reinforcement learning [17]. Each investor type is treated as an agent whose observed daily net-buy intensity reveals a latent reward defined over interpretable market-state features. A concave reward makes the optimal action graded, so trading intensity is derived rather than assumed. The three types are estimated under one *shared specification*, with only the parameters learned separately from each type's own data—so that any difference between types reflects what is learned rather than what was designed, and the estimated coefficients lie on a common scale. In this myopic formulation, the estimator coincides exactly with a Lasso (§3). We state this rather than obscure it, and build interpretation and validation on top of it: the recovered rewards are checked for out-of-sample stability under combinatorial purged cross-validation [8], for agreement with independently established behavioral regularities, and for what they fail to identify.

Our contributions are as follows.

- **A shared-specification design that places investor types on a common scale.** Where prior evidence is single-phenomenon or quarterly, estimating all three types identically from daily flow makes their weights comparable *in magnitude, not only in sign*—to our knowledge the first comparison of investor-type preferences on a common scale from daily flow.
- **A validation protocol for which recovered weights to trust.** A coefficient is interpreted only when its sign is stable across resampling and re-estimation *and* its bootstrap interval excludes zero; surviving signs are checked against independently established behavioral regularities, and everything that fails the rule—including converged but unidentified coefficients—is reported as a null rather than interpreted.
- **Empirical results on daily Korean flow.** On 973 trading days of Samsung Electronics, foreign and retail flow oppose each other on the shared signals while institutional flow shows no distinct tendency. Applied without refitting to a second stock (Hyundai Motor), the foreign and retail direction partly persists—a partial regularity, not a stock-invariant law.

On a common scale, these results reconcile what was previously fragmented: the types differ not in whether their flow persists but in the direction they continue, and aggregate institutional flow is revealed as no single behavioral unit.

The remainder of the paper is organized as follows. Section 2 reviews related work; Section 3 presents the model together with the estimation and validation methodology; Section 4 reports experiments and results; and Section 5 discusses implications, limitations, and future work, including the extension to structural reward recovery.

2 Related Work

To place our contribution, we review four lines of work and note what each leaves open.

Empirical evidence on investor-type flows. That trading behavior differs systematically across investor types is well documented across markets [1, 2, 4]. Korean foreign flow shows positive-feedback trading and daily herding [1], and short-horizon contrarian patterns appear in US individual flow [4]. The sharpest evidence comes from a Finnish register recording every investor’s daily buys, sells, and holds: Grinblatt and Keloharju [2] find contrarian behavior strongest among households, while foreign investors tend to be momentum traders. The topic remains active—Oh [13] applies detrended fluctuation analysis to Korean daily flows from 2015 to 2024 and finds long-range dependence in all three types, with the cross-type ranking visible in gross buys and sells attenuating once flows are netted. These findings, however, either characterize the time-series properties of flow or come from separate regressions on different samples and specifications, each targeting one phenomenon at a time, so the resulting coefficients cannot be placed side by side across types.

Behavioral regularities behind type differences. Three regularities recur across this literature. Medium-term return persistence underlies momentum and contrarian trading [3]. The disposition effect—reluctance to realize losses [9, 10]—is documented in US brokerage accounts and recovered from daily Finnish trades [2]. Herding measures [7] capture participants *of the same type* trading a stock simultaneously; Sias [14] separates institutions that follow each other from institutions that follow their own lagged trades. Each regularity has been established in isolation, for one phenomenon and often one investor type, so none of them indicates how strongly a given type weighs one regularity against another.

Demand systems for heterogeneous investors. A separate line estimates type-level preferences within a single framework. Kojen and Yogo [6] develop a characteristics-based demand system and estimate it on Form 13F holdings, recovering how demand varies across institution types and what that heterogeneity implies for prices. The estimation applies a shared specification to all investors, as we require, but it operates on *quarterly* holdings and on firm fundamentals rather than the behavioral state variables where the regularities above are defined. Daily flow falls outside its scope.

Inverse reinforcement learning and preference estimation in finance. Inverse reinforcement learning recovers a reward that rationalizes observed behavior, with non-uniqueness resolved by the maximum-entropy principle [17]. In finance the dominant use is learning or improving trading strategies: a sentiment-based reward learned with Gaussian process IRL to drive a trading system [16], strategy recovery under transaction costs [15], and—closest to our setting—Halperin et al. [11], who recover the implied reward of individual fund managers and feed it to a forward RL algorithm to improve their allocations. Outside finance the same machinery has been used to interpret behavior rather than to trade, for instance to characterize risk-prone and risk-averse decision making [12]. The shared objective in this line is a profitable policy or the reward of a single actor, recovered to be imitated or improved. To our knowledge, no prior work estimates the rewards of *heterogeneous investor types* on a shared feature set so that they can be compared with one another, nor subjects the recovered rewards themselves to out-of-sample stability and construct-validity testing.

3 Method

We compare investor types under a common specification. We first define one-day-ahead net-buy behaviour, then the three candidate behavioural channels and two market conditions that describe it, then a reward whose optimal action generates that behaviour, and finally the same estimator for every type. Four procedures and a feature ablation test whether the recovered signs persist across resampling schemes, forecast orderings, and estimators. We call this design shared-specification preference recovery (SPR). Its coefficients describe conditional associations in aggregate flow, not structural preferences.

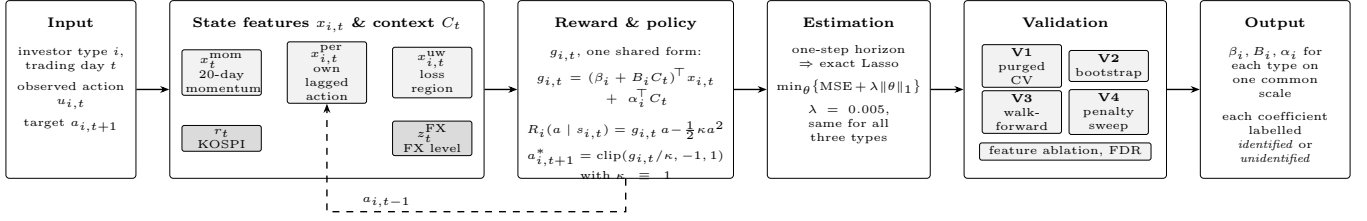


Figure 1: Shared-specification preference recovery. The three investor types receive the same three state features, the same two market contexts, the same model form, and the same penalty; only θ_i is fitted separately, from each type’s own flow. Yesterday’s action re-enters as the persistence feature (dashed). The five validation procedures examine different sources of instability, and Section 4 attributes each claim to the procedure that supports it; fitted values are reported there rather than here.

3.1 Problem setup

Let $\mathcal{I} = \{\text{foreign, institutional, retail}\}$ index investor types. For type i on day t , write $V_{i,t}^{\text{buy}}$ and $V_{i,t}^{\text{sell}}$ for gross purchase and sale value, and define the normalized net-buy action

$$u_{i,t} = \frac{V_{i,t}^{\text{buy}} - V_{i,t}^{\text{sell}}}{V_{i,t}^{\text{buy}} + V_{i,t}^{\text{sell}}} \in [-1, 1]. \quad (1)$$

Scaling net buying by the type’s own gross traded value retains both direction and intensity and places every type on the same scale regardless of its trading size. The prediction target is $a_{i,t+1} = u_{i,t+1}$, and the state $s_{i,t} = (x_{i,t}, C_t)$ contains only information observable through day t (the components of $x_{i,t}$ and C_t are defined in Section 3.2). All three types share the feature vector $x_{i,t} \in \mathbb{R}^3$, the market context $C_t \in \mathbb{R}^2$, and the model form; only the parameters are estimated separately, from each type’s own flow. Each type is treated as an agent acting independently, so no equilibrium restriction is imposed and none is recovered. This symmetry makes the fitted coefficients directly comparable across types without type-specific predictors.

3.2 State features and market context

The state vector $x_{i,t} = (x_t^{\text{mom}}, x_{i,t}^{\text{per}}, x_{i,t}^{\text{uw}})^T$ represents three candidate channels: recent price movement, the type’s own flow persistence, and an aggregate loss region. All three are specified in advance rather than selected by search; each operationalizes one of the regularities reviewed in Section 2.

Recent price movement. Let P_t be the closing price. We use the 20-day log return $x_t^{\text{mom}} = \log(P_t/P_{t-20})$ with a prespecified horizon [3]. A positive (negative) coefficient is consistent with trend-following (contrarian) behaviour [1, 2]; we expect the former for foreign flow and the latter for retail flow, with no strong prior for institutional flow.

Own flow persistence. $x_{i,t}^{\text{per}} = a_{i,t-1}$, the previous day’s action on the same own-denominator definition as (1), so the coefficient is a first-order autoregression on behaviour. Sias’s decomposition of institutional demand [14] separates institutions following each other from institutions following their own lagged trades; at the level of type aggregates those two components cannot be separated, so we use only the measurable own-following component and do not describe

this feature as herding. We expect a positive coefficient for all three types, following the long-range dependence Oh [13] reports for the netted daily flow of each type in this market, and leave the cross-type ranking open, since that paper finds the ordering does not survive netting.

Aggregate loss region. This feature approximates whether retained inventory was acquired above the current trade price. Let Q^b and Q^s be gross purchase and sale quantity, M^b and M^s the corresponding values, and $P^b = M^b/Q^b$ the purchase volume-weighted average price (VWAP) when $Q^b > 0$; type and date subscripts are suppressed and $[v]^+ = \max(v, 0)$. With $\tilde{H} = \rho H_{t-1}$,

$$O = [\tilde{H} - Q^s]^+, \quad H = [\tilde{H} + Q^b - Q^s]^+, \quad N = H - O, \quad (2)$$

$$\bar{P} = \frac{O\bar{P}_{t-1} + NP^b}{H}, \quad P^{\text{vwap}} = \frac{M^b + M^s}{Q^b + Q^s}.$$

where $\tilde{H}$ is decayed inventory entering day t , O the prior inventory surviving sales, N newly retained inventory, and $H = O + N$ total effective inventory. The recursion updates both the retained quantity and its average cost $\bar{P}$, while P^{vwap} summarizes the same-day transaction price. We initialize $H_{i,0} = 0$ and skip any VWAP term whose denominator is zero. We define the underwater feature (hence the superscript “uw”) as

$$x_{i,t}^{\text{uw}} = [(\bar{P}_{i,t} - P_{i,t}^{\text{vwap}})/\bar{P}_{i,t}]^+, \quad (3)$$

and set it to zero when effective inventory is zero or no trade occurs. The forgetting factor $\rho = 0.98$ gives recent trades greater effective weight, a half-life of roughly 34 trading days. This is an aggregate proxy rather than an account-level cost basis; its sign can be compared with disposition-effect predictions [9, 10] but cannot identify that effect. We expect a positive coefficient for retail flow, with no clear prior for the other two types.

Market conditions. Let r_t be the one-day KOSPI 200 return and e_t the won-dollar closing rate. With $\mu_{t-1,252}$ and standard deviation of $\{\log e_s : s = t - 252, \dots, t - 1\}$, define the standardized exchange-rate level and market context separately as

$$z_t^{\text{FX}} = \frac{\log e_t - \mu_{t-1,252}}{\sigma_{t-1,252}}, \quad C_t = (r_t, z_t^{\text{FX}})^\top.$$

Within each split, every feature and context variable is standardized using training indices only, and the resulting transformation is applied to the test data.

The three expectations above are fixed before estimation, so Section 4 reads the fitted coefficients against them rather than rationalizing them afterwards.

3.3 Reward and Induced Policy

For type i on day t define the scalar

$$g_{i,t} = (\beta_i + B_i C_t)^\top x_{i,t} + \alpha_i^\top C_t, \quad (4)$$

which separates an average feature sensitivity, a context-dependent adjustment of that sensitivity, and a direct context effect. Here $\beta_i \in \mathbb{R}^3$ is the sensitivity at mean context, $B_i \in \mathbb{R}^{3 \times 2}$ how that sensitivity shifts with context, and $\alpha_i \in \mathbb{R}^2$ the direct context effect when the features are at their standardized means, so $\beta_i + B_i C_t$ is the effective feature weight on day t . Writing rvec for row-major vectorization and $[\cdot; \cdot]$ for vertical concatenation,

$$w_{i,t} = [x_{i,t}; \text{rvec}(x_{i,t} C_t^\top); C_t], \quad \theta_i = [\beta_i; \text{rvec}(B_i); \alpha_i] \in \mathbb{R}^{11}, \quad (5)$$

so that $g_{i,t} = \theta_i^\top w_{i,t}$, with no intercept. We interpret the five coefficients in β_i and α_i ; the six entries of B_i are retained in every fit as conditioning terms.

Each type chooses its next-day action from the feasible set $\mathcal{A} = [-1, 1]$, the range of the normalized net-buy in (1), under the reward

$$R_i(a | s_{i,t}) = g_{i,t} a - \frac{1}{2} \kappa a^2, \quad \kappa > 0, \quad (6)$$

so that $g_{i,t}$ is the marginal reward of the first unit traded and κ the curvature of the quadratic cost. R_i is strictly concave and $\mathcal{A}$ is compact, so the optimal action exists, is unique, and is the projection of the unconstrained optimum onto $\mathcal{A}$:

$$a_{i,t+1}^* = \arg \max_{a \in \mathcal{A}} R_i(a | s_{i,t}) = \text{clip}(g_{i,t}/\kappa, -1, 1), \quad (7)$$

which we write $a_{i,t+1}^*(\theta)$ where the dependence on the parameters matters. The clip is therefore the boundary of $\mathcal{A}$ rather than a link function chosen for convenience. Only the ratio $g_{i,t}/\kappa$ is identified from observed actions, so we set $\kappa = 1$ and read $g_{i,t}$ on the action scale; magnitudes are comparable across types because every type's action lies on the common $[-1, 1]$ scale of (1). Concavity is what makes intensity informative: under a reward linear in a the optimal action would sit at a bound every day, and observed magnitudes would carry no information about the state.

Because the horizon is one step, the recovered reward is informationally equivalent to the conditional best response. We therefore read β_i as a conditional association in aggregate flow rather than as a structural preference, and do not separate preferences from beliefs or institutional constraints:

a type that buys after positive returns may prefer trend exposure, may believe returns persist, or may operate under a mandate that produces the same flow.

The predicted action is (7) evaluated at $\hat{\theta}_i$, and we fit θ_i by minimizing its squared error against the observed action under an L_1 penalty. With $\mathcal{D}_i$ the training indices for type i ,

$$\hat{\theta}_i = \arg \min_{\theta} \left\{ \frac{1}{|\mathcal{D}_i|} \sum_{t \in \mathcal{D}_i} (a_{i,t+1}^*(\theta) - a_{i,t+1})^2 + \lambda \|\theta\|_1 \right\}. \quad (8)$$

The saturation rate is zero for all three types in every split, so the bound in (7) never binds and (8) reduces to a Lasso in θ . Across all 45 splits the design matrix $W_i = [w_{i,t}^\top]_{t \in \mathcal{D}_i}$ has full column rank 11, with a worst-case condition number of 6.94, so the reduced objective is strictly convex and its minimizer is unique—which is what licenses reading the reported coefficients as a property of the estimator rather than of one solution among many.

The penalty is $\lambda = 0.005$ for every type. A common penalty is what makes coefficient magnitudes comparable: under type-specific penalties a smaller coefficient could reflect heavier shrinkage rather than weaker sensitivity.

3.4 Validation

Four procedures and a feature ablation examine different sources of instability, and Section 4 attributes each claim to the procedure that supports it.

(V1) Combinatorial purged cross-validation (CPCV).

We partition the sample into ten chronological folds and take every choice of two folds as the test set, giving $\binom{10}{2} = 45$ splits [8], with a one-day purge at test boundaries and a five-day embargo after each test block. This is the primary procedure: it supplies the sampling distribution of β_i and α_i and the common basis for the ablation and penalty comparisons. Because two test folds need not be adjacent, some splits train partly on data later than their test block, so V1 measures sensitivity to train–test composition rather than forecast skill, and we do not interpret sign consistency as a p -value.

(V2) Calendar-month block bootstrap.

Because the 45 splits share observations, sign consistency across them understates uncertainty. We resample calendar months with replacement 200 times, refitting each time, and report 95% intervals; monthly blocks preserve within-month dependence in daily flow. This procedure fits in sample and produces no held-out predictions, so we use it only for coefficient intervals.

(V3) Expanding walk-forward evaluation.

Since V1 admits splits that train on later data, we also impose strict chronology, training on all prior calendar years and testing on 2023, 2024 and 2025 in turn, with training sets of 242, 487 and 731 days. Scalers are fitted within each window's training range.

(V4) Penalty substitution.

To test dependence on the estimator we refit under an L_2 penalty over a grid of fifty strengths, select one strength per type, and compare coefficient signs with the L_1 fits.

Feature ablation. We refit the model with each feature and each context removed in turn, on the same 45 splits, and compare pooled out-of-sample performance against the full specification using a paired block bootstrap over dates with block length 20 and 10,000 resamples, controlling the false discovery rate across variants and metrics. Pooling across splits is required for the paired comparison, so these figures are not directly comparable to the per-split averages reported for V1. We report three single-feature and two context removals.

Performance is summarized by directional accuracy with $\text{sign}(0) = 0$, correlation between predicted and realized actions, MAE, RMSE, and out-of-sample R^2 against each split’s training-mean action. Because action dispersion differs across types, we do not rank types by absolute MAE.

Finally, we fix in advance what counts as identification. A coefficient is reported as identified only when its sign agrees with its CPCV mean in at least 90% of the 45 V1 splits *and* its V2 interval excludes zero; otherwise it is reported as unidentified rather than interpreted. V3 is reported alongside as evidence on temporal stability rather than on identification: with only three windows the procedure is weak in both directions. A feature whose removal does not degrade performance is reported as contributing nothing.

4 Experiments

Section 4.1 describes the source and target data and the zero-shot design. Section 4.2 reports the baseline feature weights β , direct context effects α , and feature–context interactions B estimated on Samsung Electronics. Section 4.3 freezes the Samsung-trained policies and tests whether the investor-type policies persist in Hyundai Motor. Section 4.4 summarizes the retained tendencies and their scope of interpretation.

4.1 Data and Zero-Shot Design

The daily panel for each stock contains the closing price and trading value; gross purchase and sale values and quantities for foreign, institutional, and retail investors; the one-day KOSPI 200 return; and the won–dollar exchange rate. The prediction target is the next-day normalized net-buy action for each investor type, as defined in the methodology. Gross purchase and sale quantities and values are also used to construct the aggregate average-cost proxy required by the loss-region feature.

The Samsung Electronics (005930) estimation and validation sample contains 973 state days from January 6, 2022, to December 29, 2025. The Hyundai Motor (005380) input data begin in 2021 to provide the warm-up period required for lagged features and the 252-day exchange-rate context. The resulting target sample contains 972 state days from January 7, 2022, to December 29, 2025. The one-day difference arises solely from target-stock warm-up availability.

Hyundai Motor is evaluated under a zero-shot design. For every CPCV split, the investor-specific checkpoint trained on Samsung Electronics and that split’s feature and context scalars are frozen and then applied to Hyundai Motor

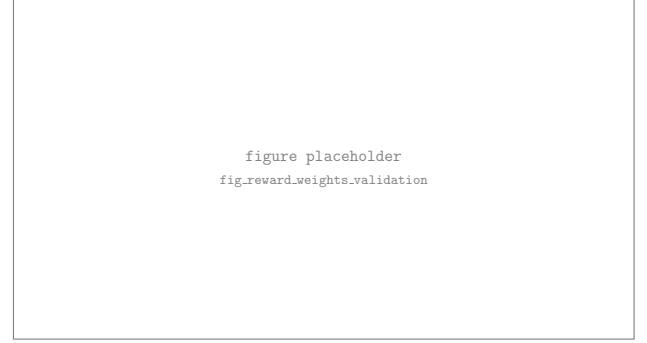


Figure 2: Baseline feature weights β for Samsung Electronics. Filled markers pass the V1+V2 stability criteria; open markers do not.

observations on the corresponding test dates. Hyundai Motor observations are never used for model fitting, scaling, hyperparameter selection, or checkpoint selection. The nine test-path predictions for each date are averaged before evaluation.

4.2 Stability of Investor Tendencies Learned from Samsung Electronics

The roles of the coefficients and the model equation are defined in the methodology. Here, we report results by distinguishing the feature sensitivities at average market context, β ; the direct context effects, α ; and B , which measures how feature sensitivities vary with market context. Table 1 reports β and α , while Table 2 presents every element of B omitted from the previous draft.

The term “identified” does not denote causal identification. It is shorthand for passing the predeclared stability criteria and therefore having an interpretable sign. A coefficient is retained for interpretation only when its sign agrees with the mean sign in at least 90% of the 45 V1 CPCV splits and the V2 calendar-month block-bootstrap 95% interval excludes zero. Table 1 reports the V2 result directly as “excludes 0” or “includes 0” and gives the final decision as “interpret” or “withhold.” V3 is an auxiliary diagnostic across three temporal windows and does not change this decision.

V1 and V2 retain five baseline feature weights for interpretation. Foreign investors exhibit positive momentum and flow persistence; retail investors exhibit negative momentum and positive flow persistence; and institutions exhibit a smaller positive flow-persistence coefficient. The four direct context effects for foreign and retail investors pass the stability criteria with opposite signs. The institutional KOSPI coefficient remains as a small negative effect, whereas its exchange-rate coefficient is withheld. Under V3, the foreign and retail momentum and persistence signs are retained in all three windows, while institutional persistence is retained in only two. Replacing L_1 with L_2 regularization preserves

Table 1: Samsung Electronics baseline feature weights β , direct context effects α , and stability decisions. β values are CPCV mean $\pm$ standard deviation, and α values are bootstrap mean [95% interval]. V1 is the share of 45 splits retaining the reported sign. V3 is the number of three expanding windows retaining that sign; V3 was not estimated for α and is reported as n/a.

Investor	Coefficient	Estimate	Stability decision
Foreign	β : momentum	$+0.0435 \pm 0.0090$	V1 100.0%; V2 excludes 0; Interpret; V3 3/3
	β : flow persistence	$+0.0505 \pm 0.0060$	V1 100.0%; V2 excludes 0; Interpret; V3 3/3
	β : loss region	$+0.0061 \pm 0.0048$	V1 91.1%; V2 includes 0; Withhold; V3 2/3
Institutional	β : momentum	-0.0015 ± 0.0022	V1 75.6%; V2 includes 0; Withhold; V3 2/3
	β : flow persistence	$+0.0102 \pm 0.0031$	V1 100.0%; V2 excludes 0; Interpret; V3 2/3
	β : loss region	-0.0003 ± 0.0006	V1 75.6%; V2 includes 0; Withhold; V3 0/3
Retail	β : momentum	-0.0356 ± 0.0066	V1 100.0%; V2 excludes 0; Interpret; V3 3/3
	β : flow persistence	$+0.0462 \pm 0.0078$	V1 100.0%; V2 excludes 0; Interpret; V3 3/3
	β : loss region	$+0.0288 \pm 0.0125$	V1 97.8%; V2 includes 0; Withhold; V3 2/3
Foreign	α : KOSPI return	$+0.0325$ [+0.0178, +0.0483]	V1 100.0%; V2 excludes 0; Interpret; V3 n/a
	α : FX level	-0.0225 [-0.0479, -0.0002]	V1 97.8%; V2 excludes 0; Interpret; V3 n/a
	α : KOSPI return	-0.0061 [-0.0119, -0.0002]	V1 100.0%; V2 excludes 0; Interpret; V3 n/a
Institutional	α : KOSPI return	-0.0038 [-0.0122, +0.0003]	V1 97.8%; V2 includes 0; Withhold; V3 n/a
	α : FX level	-0.0217 [-0.0424, -0.0059]	V1 100.0%; V2 excludes 0; Interpret; V3 n/a
	α : KOSPI return	$+0.0322$ [+0.0017, +0.0643]	V1 100.0%; V2 excludes 0; Interpret; V3 n/a

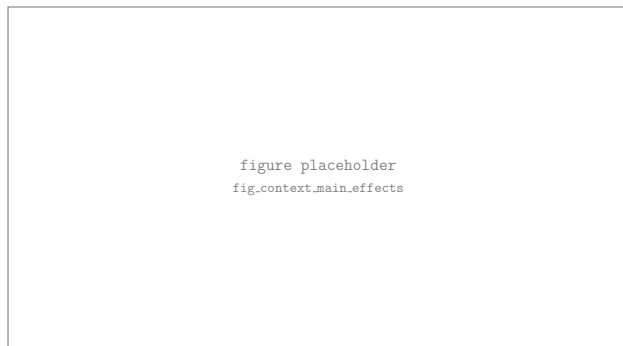


Figure 3: Direct market-context effects α for Samsung Electronics. Points are calendar-month block-bootstrap means with 95% intervals.

the signs of all nine β coefficients. Feature-removal tests support the predictive contribution of foreign flow persistence

Table 2: Feature-context interaction coefficients B for Samsung Electronics. Each cell reports the CPCV mean $\pm$ standard deviation, followed in parentheses by the share of splits retaining the mean sign. These values are reported descriptively for transparency and are not promoted to core investor tendencies based on CPCV sign consistency alone.

Investor	Feature j	$B_{j,KOSPI}$	$B_{j,FX}$
		mean $\pm$ SD (sign)	mean $\pm$ SD (sign)
Foreign	Momentum	-0.0102 ± 0.0057 (97.8%)	-0.0048 ± 0.0044 (93.3%)
	Flow persistence	-0.0029 ± 0.0046 (75.6%)	$+0.0016 \pm 0.0048$ (62.2%)
	Loss region	-0.0043 ± 0.0036 (91.1%)	-0.0042 ± 0.0052 (82.2%)
Institutional	Momentum	$+0.0003 \pm 0.0013$ (33.3%)	$+0.0014 \pm 0.0014$ (88.9%)
	Flow persistence	$+0.0004 \pm 0.0010$ (60.0%)	-0.0034 ± 0.0020 (100.0%)
	Loss region	$+0.0054 \pm 0.0021$ (100.0%)	$+0.0021 \pm 0.0015$ (95.6%)
Retail	Momentum	$+0.0056 \pm 0.0037$ (84.4%)	-0.0032 ± 0.0042 (86.7%)
	Flow persistence	$+0.0102 \pm 0.0050$ (100.0%)	-0.0041 ± 0.0075 (66.7%)
	Loss region	$+0.0033 \pm 0.0041$ (80.0%)	-0.0316 ± 0.0068 (100.0%)

Table 3: Test-path-ensemble performance for the Samsung Electronics source sample and the zero-shot Hyundai Motor target sample. Nine CPCV test-path predictions are averaged by date.

Stock	Investor	State days	Directional accuracy	Correlation	RMSE
Samsung Electronics	Foreign	973	0.642	0.344	0.242
	Institutional	973	0.534	0.056	0.126
	Retail	973	0.607	0.302	0.319
Hyundai Motor	Foreign	972	0.615	0.218	0.252
	Institutional	972	0.523	0.040	0.226
	Retail	972	0.565	0.138	0.276

but do not promote coefficients that fail V1 or V2 to interpretable results.

4.3 Do the Recovered Tendencies Transfer to Hyundai Motor?

The evaluation metrics and calculations follow the methodology. Directional accuracy is the share of dates on which the predicted and realized next-day normalized net-buy actions have the same sign, and correlation is the Pearson correlation between the two actions. RMSE measures error on the action scale. Because action dispersion differs across stocks and investor types, RMSE is reported but not used to rank behavioral retention.

The foreign and retail tendencies learned from Samsung Electronics retain some directional information in Hyundai Motor, although both weaken. Foreign directional accuracy declines from 0.642 to 0.615 and correlation from 0.344 to 0.218. Retail directional accuracy declines from 0.607 to 0.565 and correlation from 0.302 to 0.138. The institutional policy



Figure 4: Realized actions and model predictions for the Samsung Electronics source sample and the zero-shot Hyundai Motor target sample. Each line is a 20-trading-day rolling mean of the daily normalized net-buy action; black lines denote realized actions and colored lines denote model predictions.

is weak in both stocks: its Hyundai directional accuracy is 0.523 and its correlation is 0.040. Figure 4 likewise shows that the foreign and retail predictions follow part of the medium-horizon direction in realized flows, whereas institutional predictions remain near zero. The experiment therefore supports partial retention of foreign and retail tendencies, but it neither establishes a stock-invariant behavioral law nor provides meaningful evidence that the institutional tendency persists.

4.4 Interpretation and Scope

Meaning of the transfer result. The Hyundai Motor experiment does not test whether the coefficients estimated on Samsung Electronics are reproduced at the same values. It freezes the entire Samsung-trained policy and evaluates how closely that policy matches the direction of next-day Hyundai Motor flows. The positive correlations retained for foreign and retail investors therefore indicate that investor-type tendencies preserve some predictive information when the stock changes. At the same time, the decline in correlation from 0.344 to 0.218 for foreign investors and from 0.302 to 0.138 for retail investors shows that the strength and expression of those tendencies vary with stock characteristics. The decline cannot be attributed uniquely to liquidity, ownership structure, or industry, but it clearly limits any interpretation as a stock-invariant behavioral law.

Foreign investors: buying into upward trends while continuing order flow. They increase net buying as prices rise and continue the preceding trading direction.

For Samsung Electronics, foreign investors exhibit both a positive momentum coefficient and positive flow persistence.

Net buying strengthens after recent price increases, and the preceding direction of foreign net buying or selling continues. In this sense, foreign trading tends to follow rather than resist the market trend.

The direct context effects are consistent with this pattern. The KOSPI direct effect is positive and the exchange-rate direct effect is negative: net buying is stronger in rising-market states and weaker when the won is relatively weak. Foreign investors tend to enter when market conditions are favorable and retreat as exchange-rate pressure rises.

The interaction coefficients B indicate that momentum sensitivity decreases as the KOSPI return rises. This entry is only a descriptive result supported by CPCV sign consistency, however, and is presented as an auxiliary observation rather than evidence for a core tendency.

In terms of cross-stock generality, Hyundai Motor retains a directional accuracy of 0.615 and a correlation of 0.218 for the foreign policy. This suggests that the trend-following flow pattern is not purely specific to Samsung Electronics.

Retail investors: selling into rises, buying into declines, and continuing order flow. They tilt toward net selling as prices rise and continue the preceding trading direction.

The signs of the retail coefficients mirror those of foreign investors. The negative momentum coefficient means that net buying weakens or net selling strengthens after recent price increases. Positive flow persistence means that the direction formed in this way continues over the short term. Read jointly, the two coefficients describe repeated orders in the same direction across multiple periods: staggered selling during price increases and staggered buying during declines.

The direct context effects also point in the opposite direction. The KOSPI effect is negative and the exchange-rate effect is positive, consistent with retail investors absorbing flow in states in which foreign investors retreat.

The result is consistent with retail investors reducing positions as prices rise, adding to positions during declines to lower average cost, and postponing the disposal of losing positions. Retail flow can therefore be interpreted as reacting against market trends and the profit-and-loss state of holdings rather than following the market trend.

The KOSPI interaction with flow persistence and the exchange-rate interaction with the loss region suggest that response magnitudes may vary across market states. Because these estimates do not pass the complete stability decision, they are not treated as definitive conclusions. The retail correlation falls to 0.138 in Hyundai Motor, indicating that the contrarian direction partly persists but may be more stock-dependent in magnitude than the foreign tendency.

Institutional investors: no distinct common tendency is observed. Only a small positive flow-persistence coefficient and a small negative KOSPI direct effect pass the stability criteria for institutions, and predictive association remains low. The Samsung Electronics correlation is 0.056, and the persistence sign is retained in only two of the three temporal windows. In Hyundai Motor, directional accuracy is 0.523, close to chance, and correlation is only 0.040.

Two statements must therefore be separated: (i) aggregate institutional flow contains a weak persistence signal, and (ii) a distinct institutional tendency recurs in another stock. The results support only the first statement. Even when some institutional B entries in Table 2 show high CPCV sign consistency, they do not justify a strong interpretation of state-dependent institutional behavior.

This does not mean that institutional trading is random. It means that the reward features and context variables used here explain little of aggregate institutional behavior. The institutional category combines heterogeneous entities, including securities firms, insurers, investment trusts, private funds, banks, and pension funds. Index tracking, mechanical rebalancing, and hedging can also generate trades largely unrelated to the modeled price signals, so their effects may offset one another in aggregate. The conclusion is therefore that no distinct common tendency is recovered at the aggregate institutional level.

Common pattern. Flow persistence has the largest weight for all three investor groups. In the standardized model, the preceding direction of net buying or selling is the common factor most strongly reflected in next-day flow. Foreign, retail, and institutional investors thus respond more strongly to short-term order flow than to price changes or market and exchange-rate states. This pattern is consistent with orders being executed in pieces across multiple periods. The investor types differ not in whether persistence exists but in what determines the direction being continued: foreign investors follow trends, retail investors trade against them,

and aggregate institutional flow reveals no distinct common signal determining that direction.

Scope of interpretation. A coefficient describes how one variable is associated with predicted next-day net buying while the remaining features and contexts are held fixed. This is the concrete meaning of “conditional association” in this study. It is not a causal preference estimate and does not separate beliefs, delegated-management constraints, common shocks, or within-category heterogeneity. The Samsung Electronics estimates define the source-stock tendencies, and the Hyundai Motor experiment tests only whether the frozen policies preserve predictive direction in a new stock. It does not re-estimate or re-identify β , α , or B on Hyundai Motor.

5 Conclusion

This study recovers daily investor-type flow tendencies under a shared specification and tests their zero-shot transfer. On Samsung Electronics, foreign and retail investors load oppositely on momentum (+0.0435 versus −0.0356) but share positive flow persistence (+0.0505 and +0.0462). Frozen-policy evaluation on Hyundai Motor yields directional accuracies of 0.615 and 0.565 and correlations of 0.218 and 0.138, supporting partial cross-stock retention rather than a stock-invariant law; aggregate institutional flow remains weak in both stocks. Because the evidence covers only two Korean large-cap stocks and about four years of aggregate daily data, the coefficients remain conditional associations rather than causal preferences. Institutional aggregation and the average-cost loss proxy further limit interpretation. Applying the same protocol to SK hynix, NAVER, and stocks with different ownership and liquidity, together with disaggregated or account-level data, is necessary to establish broader cross-sectional scope.

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